

1. Einstein's mass-energy equation

$$E=mc^2$$

Apart from in nuclear physics, we apply the rules that mass is conserved and energy is conserved.

In nuclear physics energy can be converted into mass, and vice versa

'mass-energy' is conserved – the total amount of (mass and energy)

Rest mass (mass when at rest) – is the 'usual' meaning of mass

An object that is moving gains mass. At everyday speeds the extra amount is too small to notice. At high speeds it becomes significant

Annihilation A particle and its antiparticle, when they meet, **annihilate** - disappear, releasing energy equivalent to their total mass

Pair production Energy (in the form of a photon) can be converted to create a particle antiparticle pair, with combined mass equal to the energy.

2. Binding energy

Units

Energy can be given in joules (J) or electron volts (eV) or multiples (eg MeV)

$$1\text{eV} = 1.6 \times 10^{-19}\text{J}$$

Nucleon

Collective name for particles in a nucleus: eg if there are 2 neutrons and 3 protons, there are 5 nucleons
(It is not a new type of particle!)

Forces within the nucleus

The nucleons are held together as a nucleus by the strong nuclear force (chapter 25). This is in spite of the protons repelling each other due to electrostatic repulsion – the strong nuclear force is greater when the nucleons are this close together.

Mass defect

The mass of a nucleus is slightly less than the combined masses of the protons and neutrons it is made of.
The **mass defect** is the difference between the mass of the nucleus, and the sum of the masses of the nucleons of which it is composed

Binding energy

$$\text{Binding energy} = \text{mass defect} \times c^2$$

Binding energy is the minimum energy needed to completely separate a nucleus into its constituent nucleons

Binding energy per nucleon

Binding energy of a nucleus divided by the number of nucleons.

You can think of it as the energy needed to remove one nucleon (it is actually the mean)

A high binding energy. per nucleon indicates a nucleus where the nucleons are tightly bound (by the strong nuclear force)

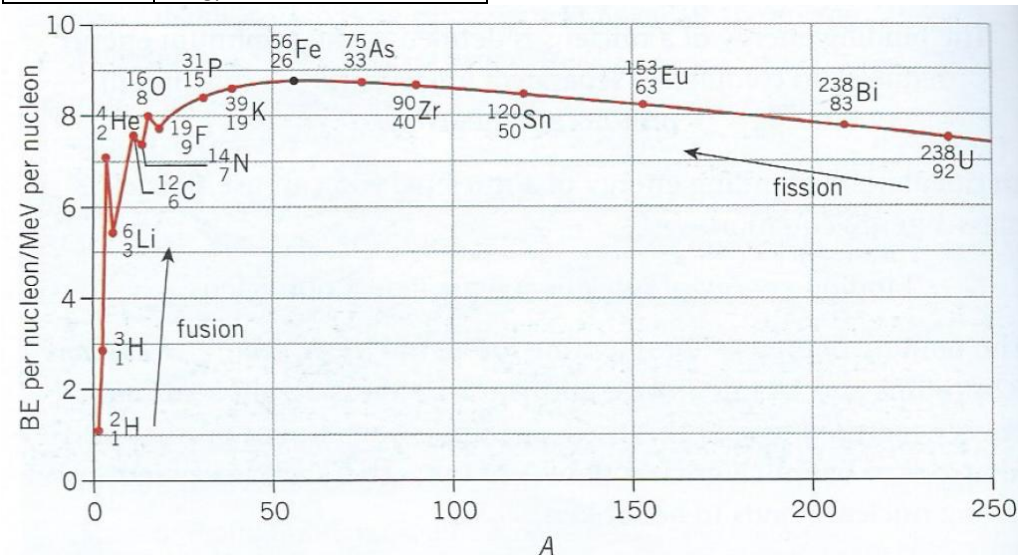
Graph

The y-axis is binding **energy per nucleon**, not binding energy

The peak of the curve is Fe – the most stable nucleus

Nuclei on the left of Fe release energy in fusion

Nuclei on the right of Fe release energy in fission

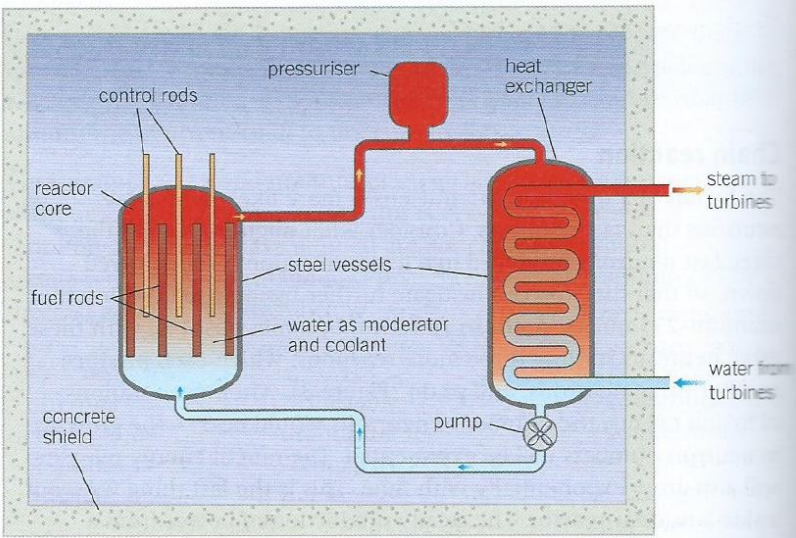
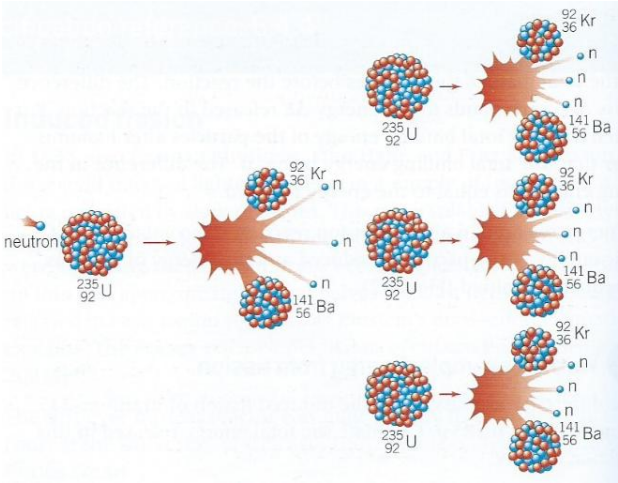


3. Nuclear fission	
Nuclear fission	The breaking apart / splitting up of large nuclei to release energy
Nuclear power	Fission is the process used in nuclear power stations.
Uranium as fuel	Uranium releases energy through the process of nuclear fission. Uranium described as fuel, but it is important that this is not combustion – it is not a chemical reaction at all.
Induced fission	A uranium-235 nucleus absorbs a neutron, becomes U-236 which is highly unstable and breaks into two daughter nuclei, and some neutrons
Thermal neutrons	Slow moving neutrons. Only these thermal neutrons can be absorbed and set off the fission process.
3 neutrons released	Typically 3 neutrons are released in this process; they have high energy so are moving at high speed.

Chain reaction	The three neutrons released can be absorbed by 3 other nuclei, causing them to fission, and those 3 can each trigger three more – this exponential growth is called a chain reaction. It can only happen in specific circumstances, including that the three neutrons are slowed down to become thermal neutrons
Waste products – advantages	No CO ₂ is released; this is not a combustion reaction. This is a significant advantage for fission, compared with fossil fuel power stations
Waste products - disadvantages	The uranium decays into a large number of other elements, some of which are highly radioactive, eg plutonium. This waste must be carefully managed for many years.

Control rods	Absorb neutrons; can be raised (to allow the chain reaction) or lowered (to stop it, or slow it) Made of boron
Moderator	To slow the neutrons; they must become thermal neutrons for the chain reaction to happen. Graphite and water are both good moderators.
Coolant	Carries the thermal energy released to a heat exchanger, and on to a turbine to generate electricity. Despite the name its purpose is to transfer the energy, not to cool the reactor. Water is a good coolant; in some designs the water acts as both coolant and moderator

Fusion	The joining together of light nuclei to release energy
	Occurs in the sun due to the high temperature and pressure
Hydrogen and helium.	Two hydrogen nuclei fusing together to produce a helium nucleus is a typical example.
Conditions	The nuclei have positive charges, so require a combination of high temperature/pressure/density to bring them close enough together to fuse
As an energy resource	A nuclear fusion power station would be ideal in many ways: hydrogen is abundant, the waste product helium is harmless (and valuable), and a huge amount of energy is released.
Technology	Designs for a fusion reactor centre on ways to force hydrogen nuclei together, usually by accelerating them to high speeds and/or containing them using magnetic fields. This has been achieved but not yet in an economically viable way (it is too expensive to be useful)
20 years	Physicists confidently expect fusion power stations to be working within 20 years. We have been saying that for at least 20 years, but we're <i>really</i> confident now.



4. Nuclear fusion